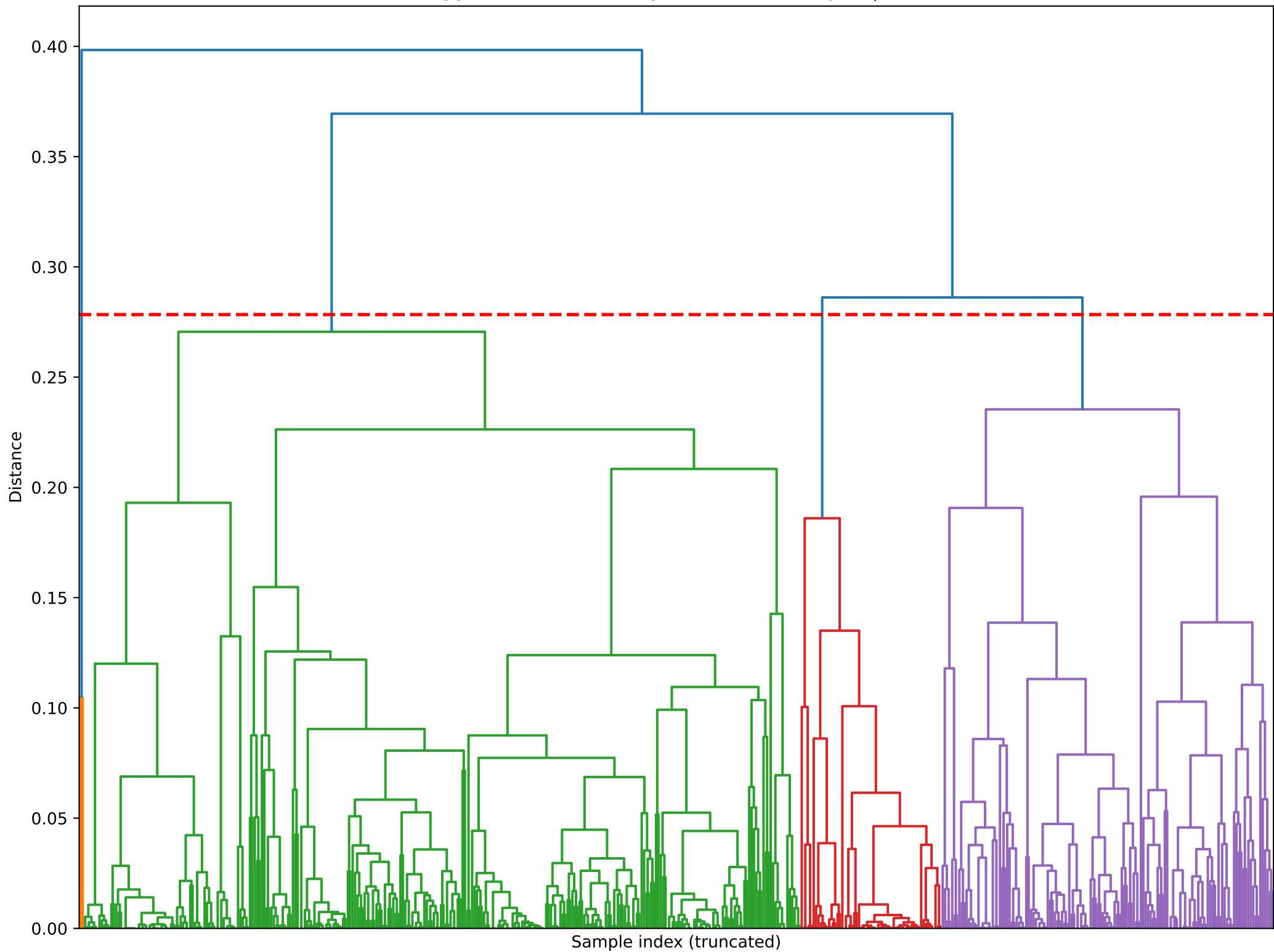


Introduction — experimental setup

Pipeline overview (action → purpose):

- 1) Load `5\_test\_candidate\_cleaned\_final.csv` and validate key column names (e.g., EXAM_SUBJECT, CREDENTIAL_DATE, SCORE_MISSING) → ensure the Python pipeline matches the R reference inputs.
- 2) Aggregate exam-attempt rows to one row per `CANDIDATE\_ID` (scores, pass>=145, college>=165, avg_retakes formula; subject means only if attempted; missing subjects stay NA) → create a candidate-level clustering dataset.
- 3) Derive profiling-only fields (e.g., age, census_region, race_category and *_label) → support interpretation tables without affecting clusters.
- 4) Completeness Stage 1: treat sentinel 0 as missing for demographics; compute completeness across 26 key fields; keep candidates with completeness ≥ 0.70 → remove records with too little usable information.
- 5) Completeness Stage 2: drop candidates with no demographic identity (no gender AND no race AND no ethnicity) → remove un-profileable cases.
- 6) Variance filter on numeric clustering columns (relative variance ≥ 0.15 ; protect `avg\_score`) → drop near-constant features with no segmentation power.
- 7) Correlation filter on retained numeric columns ($|r| > 0.80$ drop one; protect `avg\_score`) → reduce redundancy / multicollinearity.
- 8) Preprocess for mixed data: continuous vs binary; continuous z-score; non-subject NA → 0, subject NA kept; binary stays 0/1 → align scales while preserving missingness semantics.
- 9) Compute Gower distance on clustering features only (exclude IDs and all demographic/geography) → measure dissimilarity for mixed types.
- 10) Run agglomerative hierarchical clustering (average linkage); choose k via silhouette (2..8) and dendrogram → select an interpretable, data-supported partition.
- 11) Output cluster labels and post-hoc tables (performance, subject, preparation, demographics) → describe cluster profiles after clustering.
- 12) Scope decisions: do NOT implement DIANA; do NOT use study/reason/enrollment as clustering features (profiling only) → avoid mixing survey behavior fields into segmentation.

Agglomerative dendrogram (n=500 sample) | k=4



Using SELECTED_K and final cluster sizes

Using SELECTED_K: 4

Final cluster sizes:
cluster

1	415
2	1118
3	1667
4	497

Name: count, dtype: int64

Post-hoc summary — Performance (clustering features)

cluster	N	Avg_Score	Pass_Rate	College_Ready	Credential_Earned	Avg_Retakes	Pct_Online	Used_GED_Ready	Has_Prep_Center
1	415	157.1162	93.8%	19.9%	100.0%	0.1024	0.0%	0.0%	12.3%
2	1118	155.6158	88.4%	18.2%	100.0%	1.3630	13.7%	100.0%	37.8%
3	1667	146.2461	53.2%	6.3%	0.0%	0.8635	4.9%	100.0%	42.8%
4	497	145.0541	39.4%	3.9%	0.0%	0.2181	0.0%	0.0%	20.3%

Post-hoc summary — Subject scores

cluster	score_math	score_science	score_reasoning	score_social_studies
1	154.9	158.5	157.6	158.7
2	153.3	157.9	156.1	156.9
3	144.2	149.3	146.2	147.4
4	142.0	146.2	145.4	145.5

Post-hoc summary — Preparation behaviors

cluster	studied_for_ged	study_helpful_adult_education_class	study_helpful_online_course_video_study_materials
1	79.5%	24.1%	22.4%
2	81.9%	28.6%	25.7%
3	77.6%	26.8%	19.4%
4	81.7%	36.0%	13.5%

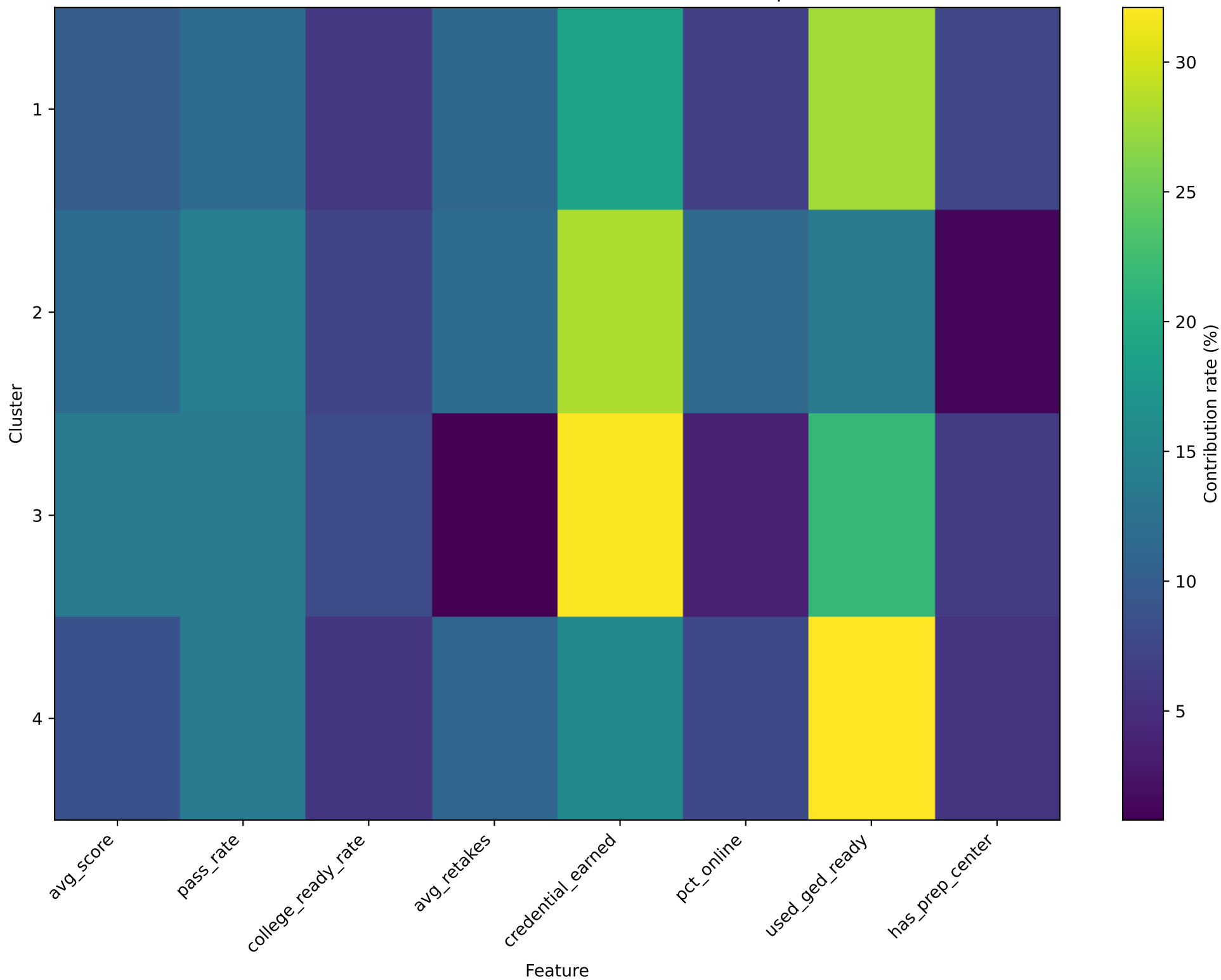
Post-hoc summary — Demographics

cluster	Avg_Age	Male	Female	Black	White	Hispanic
1	27.6651	67.0%	26.3%	17.1%	57.1%	13.7%
2	27.6941	51.5%	39.5%	13.7%	59.8%	19.0%
3	30.7510	44.5%	48.7%	23.2%	47.2%	24.8%
4	31.9215	56.1%	38.8%	33.6%	33.2%	23.1%

Cluster × Feature contribution matrix (%)

	avg_score	pass_rate	college_ready_rate	avg_retakes	credential_earned	pct_online	used_ged_ready	has_prep_center
1	10.0	11.8	6.0	11.3	18.9	6.7	27.8	7.5
2	11.7	14.2	7.3	11.9	28.2	11.6	13.6	1.4
3	13.8	13.8	7.9	0.8	31.9	3.7	21.7	6.4
4	8.7	13.6	5.7	11.0	15.5	7.7	32.1	5.6

Cluster x Feature contribution rates (heatmap)



Top features per cluster (by contribution %)

=== Top features per cluster (by contribution %) ===

- Cluster 1: used_ged_ready (27.8%), credential_earned (18.9%), pass_rate (11.8%), avg_retakes (11.3%), avg_score (10.0%)
- Cluster 2: credential_earned (28.2%), pass_rate (14.2%), used_ged_ready (13.6%), avg_retakes (11.9%), avg_score (11.7%)
- Cluster 3: credential_earned (31.9%), used_ged_ready (21.7%), avg_score (13.8%), pass_rate (13.8%), college_ready_rate (7.9%)
- Cluster 4: used_ged_ready (32.1%), credential_earned (15.5%), pass_rate (13.6%), avg_retakes (11.0%), avg_score (8.7%)

Cluster 1: High-performing, credentialed candidates with low use of support resources. They tended to pass quickly, had very few retakes, and rarely used GED Ready or online testing.

Cluster 2: High-performing, credentialed candidates with high use of support resources. Their outcomes were still strong, but they relied more on GED Ready, prep centers, and repeated attempts.

Cluster 3: Lower-performing, non-credentialed candidates with high support usage. They frequently used GED Ready and prep centers, but these supports did not fully translate into credential completion.

Cluster 4: Lowest-performing, non-credentialed candidates with low support usage. They showed weak outcomes, little use of preparation resources, and low retake intensity, making them the most at-risk group.